9.2 Motion on a Curve

Introduction In the preceding section we saw that the first and second derivatives of the vector function $\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ can be obtained by differentiating the component functions f, g, and h. In this section we give a physical interpretation to the vectors $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

Velocity and Acceleration Suppose a body or particle moves along a curve C so that its position at time t is given by the vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. If the component functions f, g, and h have second derivatives, then the vectors

$$\mathbf{v}(t) = \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$$

$$\mathbf{a}(t) = \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}$$

are called the **velocity** and **acceleration** of the particle, respectively. The scalar function $\|\mathbf{v}(t)\|$ is the **speed** of the particle. Since

$$\|\mathbf{v}(t)\| = \left\|\frac{d\mathbf{r}}{dt}\right\| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

speed is related to arc length s by $s'(t) = \|\mathbf{v}(t)\|$. In other words, arc length is given by $s = \int_{t_0}^{t_1} \|\mathbf{v}(t)\|\, dt$. It also follows from the discussion of Section 9.1 that if $P(x_1, y_1, z_1)$ is the position of the particle on C at time t_1, then we may draw $\mathbf{v}(t_1)$ *tangent to C at P*. Similar remarks hold for curves traced by the vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$.

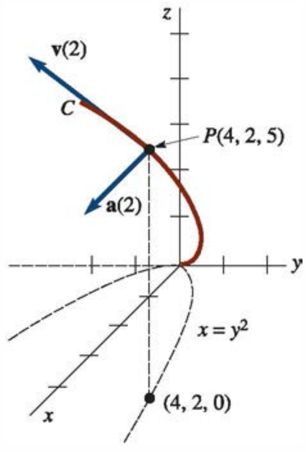


FIGURE 9.2.1 Velocity and acceleration vectors in Example 1

EXAMPLE 1 Velocity and Acceleration Vectors

The position of a moving particle is given by $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j} + \frac{5}{2}t\mathbf{k}$. Graph the curve defined by $\mathbf{r}(t)$ and the vectors $\mathbf{v}(2)$ and $\mathbf{a}(2)$.

SOLUTION Since $x = t^2$, $y = t$, the path of the particle is above the parabola $x = y^2$. When $t = 2$ the position vector $\mathbf{r}(2) = 4\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ indicates that the particle is at the point $P(4, 2, 5)$. Now,

$$\mathbf{v}(t) = \mathbf{r}'(t) = 2t\,\mathbf{i} + \mathbf{j} + \frac{5}{2}\mathbf{k} \quad \text{and} \quad \mathbf{a}(t) = \mathbf{r}''(t) = 2\mathbf{i}$$

so that $\mathbf{v}(2) = 4\mathbf{i} + \mathbf{j} + \frac{5}{2}\mathbf{k}$ and $\mathbf{a}(2) = 2\mathbf{i}$. These vectors are shown in **FIGURE 9.2.1**. ≡

If a particle moves with a constant speed c, then its acceleration vector is perpendicular to the velocity vector $\mathbf{v}$. To see this, note that $\|\mathbf{v}\|^2 = c^2$ or $\mathbf{v} \cdot \mathbf{v} = c^2$. We differentiate both sides with respect to t and obtain, with the aid of Theorem 9.1.4(*iii*):

$$\frac{d}{dt}(\mathbf{v} \cdot \mathbf{v}) = \mathbf{v} \cdot \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = 2\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = 0.$$

Thus, $$\frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = 0 \quad \text{or} \quad \mathbf{a}(t) \cdot \mathbf{v}(t) = 0 \ \text{ for all } t.$$

EXAMPLE 2 Velocity and Acceleration Vectors

Suppose the vector function in Example 2 of Section 9.1 represents the position of a particle moving in a circular orbit. Graph the velocity and acceleration vector at $t = \pi/4$.

SOLUTION Recall that $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + 3\mathbf{k}$ is the position vector of a particle moving in a circular orbit of radius 2 in the plane $z = 3$. When $t = \pi/4$ the particle is at the point $P(\sqrt{2}, \sqrt{2}, 3)$. Now,

$$\mathbf{v}(t) = \mathbf{r}'(t) = -2\sin t\mathbf{i} + 2\cos t\mathbf{j}$$

$$\mathbf{a}(t) = \mathbf{r}''(t) = -2\cos t\mathbf{i} - 2\sin t\mathbf{j}.$$

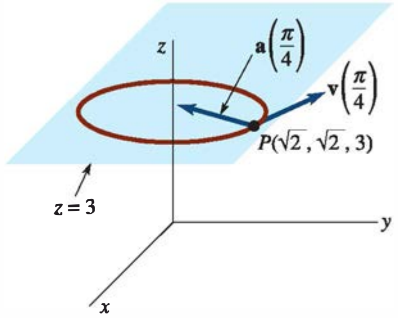


FIGURE 9.2.2 Velocity and acceleration vectors in Example 2

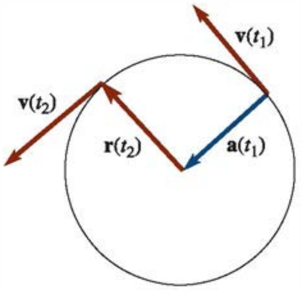


FIGURE 9.2.3 Circular motion

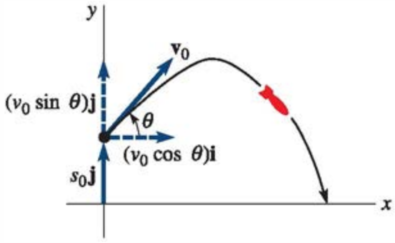
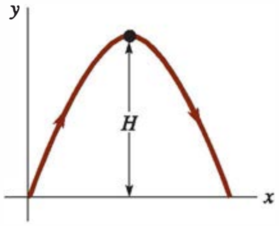
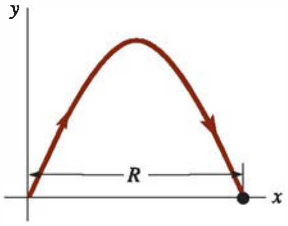


FIGURE 9.2.4 Trajectory of a projectile



(a) **Maximum height H:** Find t_1 for which $y'(t_1) = 0$; $H = y_{max} = y(t_1)$



(b) **Range R:** Find $t_1 > 0$ for which $y(t_1) = 0$; $R = x_{max} = x(t_1)$

FIGURE 9.2.5 Maximum height and range of a projectile

Since the speed is $\|\mathbf{v}(t)\| = 2$ for all time t, it follows from the discussion preceding this example that $\mathbf{a}(t)$ is perpendicular to $\mathbf{v}(t)$. (Verify this.) As shown in **FIGURE 9.2.2**, the vectors

$$\mathbf{v}\left(\frac{\pi}{4}\right) = -2\sin\frac{\pi}{4}\mathbf{i} + 2\cos\frac{\pi}{4}\mathbf{j} = -\sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j}$$

$$\mathbf{a}\left(\frac{\pi}{4}\right) = -2\cos\frac{\pi}{4}\mathbf{i} - 2\sin\frac{\pi}{4}\mathbf{j} = -\sqrt{2}\mathbf{i} - \sqrt{2}\mathbf{j}$$

are drawn at the point P. The vector $\mathbf{v}(\pi/4)$ is tangent to the circular path and $\mathbf{a}(\pi/4)$ points along a radius toward the center of the circle. ≡

Centripetal Acceleration

For circular motion in the plane, described by $\mathbf{r}(t) = r_0 \cos \omega t\mathbf{i} + r_0 \sin \omega t\mathbf{j}$, r_0 and ω constants, it is evident that $\mathbf{r}'' = -\omega^2\mathbf{r}$. This means that the acceleration vector $\mathbf{a}(t) = \mathbf{r}''(t)$ points in the direction opposite to that of the position vector $\mathbf{r}(t)$. We then say $\mathbf{a}(t)$ is **centripetal acceleration**. See **FIGURE 9.2.3**. If $v = \|\mathbf{v}(t)\|$ and $a = \|\mathbf{a}(t)\|$, we leave it as an exercise to show that $a = v^2/r_0$.

Curvilinear Motion in the Plane

Many important applications of vector functions occur in regard to curvilinear motion in a plane. For example, planetary and projectile motion take place in a plane.

In analyzing the motion of short-range ballistic projectiles,* we begin with the acceleration of gravity written in vector form: $\mathbf{a}(t) = -g\mathbf{j}$.

If, as shown in **FIGURE 9.2.4**, a projectile is launched with an initial velocity $\mathbf{v}_0 = v_0 \cos\theta\mathbf{i} + v_0 \sin\theta\mathbf{j}$ from an initial height $\mathbf{s}_0 = s_0\mathbf{j}$, then

$$\mathbf{v}(t) = \int (-g\mathbf{j})\,dt = -gt\mathbf{j} + \mathbf{c}_1,$$

where $\mathbf{v}(0) = \mathbf{v}_0$ implies that $\mathbf{c}_1 = \mathbf{v}_0$. Therefore,

$$\mathbf{v}(t) = (v_0\cos\theta)\mathbf{i} + (-gt + v_0\sin\theta)\mathbf{j}.$$

Integrating again and using $\mathbf{r}(0) = \mathbf{s}_0$ yields

$$\mathbf{r}(t) = (v_0\cos\theta)t\mathbf{i} + \left[-\frac{1}{2}gt^2 + (v_0\sin\theta)t + s_0\right]\mathbf{j}. \tag{1}$$

Hence, parametric equations for the trajectory of the projectile are

$$x(t) = (v_0\cos\theta)t, \qquad y(t) = -\frac{1}{2}gt^2 + (v_0\sin\theta)t + s_0. \tag{2}$$

We are naturally interested in finding the maximum height H and the range R attained by a projectile. As shown in **FIGURE 9.2.5**, these quantities are the maximum values of $y(t)$ and $x(t)$, respectively.

EXAMPLE 3 Trajectory of a Projectile

A projectile is launched from ground level with an initial speed $v_0 = 768$ ft/s at an angle of elevation $\theta = 30°$. Find **(a)** the vector function and parametric equations of the projectile's trajectory, **(b)** the maximum altitude attained, **(c)** the range of the projectile, and **(d)** the impact speed.

SOLUTION **(a)** The initial height and velocity are, respectively, $\mathbf{s}_0 = \mathbf{0}$ and

$$\mathbf{r}'(0) = \mathbf{v}_0 = (768\cos 30°)\mathbf{i} + (768\sin 30°)\mathbf{j} = 384\sqrt{3}\mathbf{i} + 384\mathbf{j}. \tag{3}$$

Integrating $\mathbf{a}(t) = -32\mathbf{j}$ and using (3) give

$$\mathbf{v}(t) = (384\sqrt{3})\mathbf{i} + (-32t + 384)\mathbf{j}. \tag{4}$$

*A projectile is shot or hurled rather than self-propelled. In the analysis of *long-range* ballistic motion, the curvature of the Earth must be taken into consideration.

Integrating (4) and using $\mathbf{r}(0) = \mathbf{s}_0 = \mathbf{0}$ then give

$$\mathbf{r}(t) = (384\sqrt{3})t\,\mathbf{i} + (-16t^2 + 384t)\mathbf{j}.$$

The components of this vector function,

$$x(t) = (384\sqrt{3})t, \quad y(t) = -16t^2 + 384t. \tag{5}$$

are the parametric equations of the projectile's trajectory.

(b) From (5) we see that $y'(t) = 0$ when $-32t + 384 = 0$ or $t = 12$ s. The maximum height attained by the projectile is

$$H = y(12) = -16(12)^2 + 384(12) = 2304 \text{ ft}.$$

(c) From (5) we see that $y(t) = 0$ when $-16t^2 + 384t = -16t(t - 24) = 0$. The time that projectile hits the ground is $t = 24$ s and the corresponding range is

$$R = x(24) = 384\sqrt{3}(24) \approx 15{,}963 \text{ ft}.$$

(d) Finally, from (4) we see that $\mathbf{v}(24) = (384\sqrt{3})\mathbf{i} + (-384)\mathbf{j}$ and so the impact speed is

$$\|\mathbf{v}(24)\| = \sqrt{(384\sqrt{3})^2 + (-384)^2} = 768 \text{ ft/s}. \qquad \equiv$$

In Example 3, note that the impact speed is the same as the launch speed $v_0 = 768$ ft/s. Verify that this is still the case if we change the angle of elevation to, say, $\theta = 50°$. See Problem 16 in Exercises 9.2.

Remarks

We have seen that the rate of change of arc length ds/dt is the same as the speed $\|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|$. However, as we shall see in the next section, it does *not* follow that the *scalar acceleration* d^2s/dt^2 is the same as $\|\mathbf{a}(t)\| = \|\mathbf{r}''(t)\|$. See Problem 24 in Exercises 9.2.

9.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-21.

In Problems 1–8, $\mathbf{r}(t)$ is the position vector of a moving particle. Graph the curve and the velocity and acceleration vectors at the indicated time. Find the speed at that time.

1. $\mathbf{r}(t) = t^2\mathbf{i} + \frac{1}{4}t^4\mathbf{j}; \; t = 1$
2. $\mathbf{r}(t) = t^2\mathbf{i} + \frac{1}{t^2}\mathbf{j}; \; t = 1$
3. $\mathbf{r}(t) = -\cosh 2t\,\mathbf{i} + \sinh 2t\,\mathbf{j}; \; t = 0$
4. $\mathbf{r}(t) = 2\cos t\,\mathbf{i} + (1 + \sin t)\mathbf{j}; \; t = \pi/3$
5. $\mathbf{r}(t) = 2\mathbf{i} + (t-1)^2\mathbf{j} + t\mathbf{k}; \; t = 2$
6. $\mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + t^3\mathbf{k}; \; t = 2$
7. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}; \; t = 1$
8. $\mathbf{r}(t) = t\mathbf{i} + t^3\mathbf{j} + t\mathbf{k}; \; t = 1$
9. Suppose $\mathbf{r}(t) = t^2\mathbf{i} + (t^3 - 2t)\mathbf{j} + (t^2 - 5t)\mathbf{k}$ is the position vector of a moving particle. At what points does the particle pass through the xy-plane? What are its velocity and acceleration at these points?
10. Suppose a particle moves in space so that $\mathbf{a}(t) = \mathbf{0}$ for all time t. Describe its path.
11. A shell is fired from ground level with an initial speed of 480 ft/s at an angle of elevation of 30°. Find:
    - **(a)** a vector function and parametric equations of the shell's trajectory,
    - **(b)** the maximum altitude attained,
    - **(c)** the range of the shell, and
    - **(d)** the speed at impact.
12. Rework Problem 11 if the shell is fired with the same initial speed and the same angle of elevation but from a cliff 1600 ft high.

13. A used car is pushed off an 81-ft-high sheer seaside cliff with a speed of 4 ft/s. Find the speed at which the car hits the water.

14. A small projectile is launched from ground level with an initial speed of 98 m/s. Find the possible angles of elevation so that its range is 490 m.

15. A football quarterback throws a 100-yd "bomb" at an angle of 45° from the horizontal. What is the initial speed of the football at the point of release?

16. If a projectile is launched from level ground, then the initial conditions are $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = \mathbf{v}_0 = (v_0 \cos\theta)\mathbf{i} + (v_0 \sin\theta)\mathbf{j}$ and the vector function (1) becomes

$$\mathbf{r}(t) = (v_0 \cos\theta)t\mathbf{i} + \left[-\tfrac{1}{2}gt^2 + (v_0 \sin\theta)t\right]\mathbf{j}. \qquad (6)$$

This function is equivalent to the solution $x(t)$, $y(t)$ of the system of linear differential equations (8) in Problem 21 of Exercises 4.6 subject to the same initial conditions. Underlying that earlier discussion, as well as in the derivation of (1), is the assumption that the projectile is not subject to air resistance. Use (6) to show that the impact speed of a projectile is the same as the initial speed $\|\mathbf{r}'(0)\| = \|\mathbf{v}_0\| = v_0$ for any angle of elevation θ, $0° < \theta < 90°$.

17. When air resistance is ignored, we saw in Problem 21 of Exercises 4.6 that a projectile launched at an angle θ, $0° < \theta < 90°$ from level ground ($\mathbf{r}(0) = \mathbf{0}$), then its horizontal range and maximum height are, respectively,

$$R = \frac{v_0^2}{g}\sin 2\theta \qquad \text{and} \qquad H = \frac{v_0^2}{2g}\sin^2\theta.$$

From the first formula, it follows that when the projectile is launched at distinct complementary angles the horizontal range is the same and that the maximum range is attained when the angle of elevation is $\theta = 45°$. But if the projectile is launched from an initial height $s_0 > 0$ the foregoing formulas and statements are not valid.

(a) With $v_0 = 480$ ft/s, $s_0 = 1600$ ft rework Problem 12, but this time use the complementary angle of elevation $\theta = 60°$. Compare the maximum height, range, and impact speed of the projectile with the answers to (b), (c) and (d) of Problem 12.

(b) Use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$ in part (a) of Problem 12. Repeat for the parametric equations in part (a) of this problem. Superimpose both of these curves on the same coordinate system.

18. As mentioned in Problem 17, if a projectile is launched from an initial height $s_0 > 0$ the maximum range of the projectile is not attained using $\theta = 45°$ as the angle of elevation.

(a) A projectile is launched from an initial height of $v_0 = 480$ ft/s, $s_0 = 1600$ ft, at an angle of elevation of $\theta = 45°$. Use a calculator or CAS to find the time the projectile hits the ground and the corresponding range.

(b) If the angle of elevation in part (a) is changed to $\theta = 39.76°$, show that the range is greater than that in part (a).

(c) Use a graphing utility or CAS to plot the trajectories of the projectiles in parts (a) and (b). Superimpose both of these curves on the same coordinate system.

19. If a projectile is launched from level ground, the initial conditions are $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = \mathbf{v}_0 = (v_0 \cos\theta)\mathbf{i} + (v_0 \sin\theta)\mathbf{j}$, and if linear air resistance is taken into consideration, the vector function analogue of (1) is

$$\mathbf{r}(t) = \frac{mv_0\cos\theta}{\beta}(1 - e^{-\beta t/m})\mathbf{i} + \left[\left(\frac{mv_0\sin\theta}{\beta} + \frac{m^2 g}{\beta^2}\right)(1 - e^{-\beta t/m}) - \frac{mg}{\beta}t\right]\mathbf{j},$$

This function is equivalent to the solution $x(t)$, $y(t)$ of the system of linear differential equations (11) in Problem 22 of Exercises 4.6 subject to the same initial conditions. Here $\beta > 0$ is a constant of proportionality related to the air resistance or drag. If $m = \frac{1}{4}$ slug, $g = 32$ ft/s², $\beta = 0.02$, $v_0 = 300$ ft/s, and $\theta = 38°$, use a calculator or CAS to find the impact speed of the projectile. See part (b) of Problem 22 in Exercises 4.6.

20. Suppose the angle of elevation of the projectile in Problem 19 is changed to $\theta = 52°$. Do you think that the impact speed of the projectile is the same, greater than, or less than the value found in that problem? Prove your assertion.

21. A projectile is fired from a cannon directly at a target that is dropped from rest simultaneously as the cannon is fired. Show that the projectile will strike the target in midair. See **FIGURE 9.2.6**. [*Hint*: Assume that the origin is at the muzzle of the cannon and that the angle of elevation is θ. If $\mathbf{r}_p$ and $\mathbf{r}_t$ are position vectors of the projectile and target, respectively, is there a time at which $\mathbf{r}_p = \mathbf{r}_t$?]

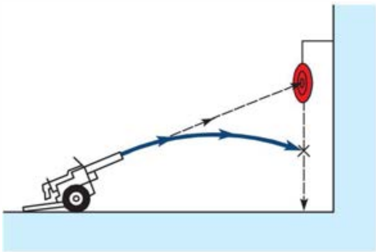


FIGURE 9.2.6 Cannon in Problem 21

22. In army field maneuvers sturdy equipment and supply packs are simply dropped from planes that fly horizontally at a slow speed and a low altitude. A supply plane flies horizontally over a target at an altitude of 1024 ft at a constant speed of 180 mi/h. Use (1) to determine the horizontal distance a supply pack travels relative to the point from which it was dropped. At what line-of-sight angle α should the supply pack be released in order to hit the target indicated in **FIGURE 9.2.7**?

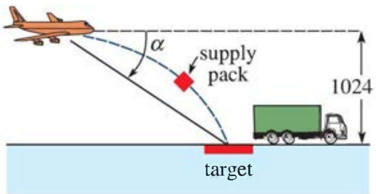


FIGURE 9.2.7 Supply plane in Problem 22